

$$E \{ \underline{y}(k) \mid \mathcal{I}(k-1) \} = E \left\{ G(k) \underline{x}(k) + \underline{\eta}(k) \mid \mathcal{I}(k-1) = \left\{ \begin{matrix} \underline{y}(0) \dots \underline{y}(k-1) \\ \underline{u}(0) \dots \underline{u}(k-1) \end{matrix} \right\} \right\} = \quad (191)$$

$$= G(k) E \{ \underline{x}(k) \mid \mathcal{I}(k-1) \} + E \{ \underline{\eta}(k) \mid \mathcal{I}(k-1) \} =$$

↑ uncorrelated ↑

$$= G(k) \hat{\underline{x}}(k|k-1) + E \{ \underline{\eta}(k) \}$$

$$= G(k) \hat{\underline{x}}(k|k-1) + \underline{0}.$$

$$\hat{\underline{y}}(k|k-1) = G(k) \hat{\underline{x}}(k|k-1)$$

Note: $\hat{\underline{y}}(k|k-1) \triangleq E \{ \underline{y}(k) \mid \mathcal{I}(k-1) \}$

$$S(k) \triangleq \text{cov} \{ \underline{y}(k) \mid \mathcal{I}(k-1) \} = E \{ [\underline{y}(k) - \hat{\underline{y}}(k|k-1)] [\underline{y}(k) - \hat{\underline{y}}(k|k-1)]^T \mid \mathcal{I}(k-1) \} =$$

$$= E \left\{ [G(k) \underline{x}(k) + \underline{\eta}(k) - G(k) \hat{\underline{x}}(k|k-1)] [\dots]^T \mid \mathcal{I}(k-1) \right\} =$$

↑ same ↑

$$= E \left\{ [G(k) [\underline{x}(k) - \hat{\underline{x}}(k|k-1)] + \underline{\eta}(k)] [\dots]^T \mid \mathcal{I}(k-1) \right\} =$$

$$= G(k) \underbrace{E \{ [\underline{x}(k) - \hat{\underline{x}}(k|k-1)] [\dots]^T \mid \mathcal{I}(k-1) \}}_{\Sigma(k|k-1)} G(k)^T +$$

$$+ G(k) E \{ [\underline{x}(k) - \hat{\underline{x}}(k|k-1)] \underline{\eta}(k)^T \mid \mathcal{I}(k-1) \} + \dots \quad (= 0)$$

uncorr $\Rightarrow E \{ \text{product} \} = \text{prod. } E \{ \cdot \}, E \{ \underline{\eta}(k) \} = 0$

$$+ E \{ \underline{y}(k) [\underline{x}(k) - \hat{\underline{x}}(k|k-1)]^T \underline{a}(k)^T | \mathcal{I}(k-1) \} + \quad (=0) \quad (1.92)$$

$$+ E \{ [\underline{y}(k) - 0] [\underline{y}(k) - 0]^T | \mathcal{I}(k-1) \} =$$

$$\boxed{S(k) = \underline{a}(k) \Sigma(k|k-1) \underline{a}(k)^T + \underline{R}(k)}$$

$\nwarrow \nearrow$
~~uncorr.~~

Note: ~~we denote~~ $S(k)$ Note: we denote ~~$\Sigma(k|k)$~~ $\text{cov} \{ \underline{y}(k) | \mathcal{I}(k-1) \}$ as $S(k)$ and not $S(k|k-1)$

In fact $S(k|k)$ would not make sense because $\text{cov}(\underline{y}(k) | \mathcal{I}(k) = \{ \underline{y}(0) \dots \underline{y}(k) \}) = 0$

end $\hat{\underline{y}}(k|k) = \underline{y}(k) !!!$

$\underline{y}(k)$ is deterministic as $\underline{y}(k)$ is available.

It can be demonstrated (see Bar Shalom et al)

$$\boxed{\hat{\underline{x}}(k|k) \triangleq E \{ \underline{x}(k) | \mathcal{I}(k) \} = \hat{\underline{x}}(k|k-1) + H(k) [\underline{y}(k) - \hat{\underline{y}}(k|k-1)]}$$

with

$$H(k) = \sum (k|k-1) G(k)^T S(k)^{-1}$$

and

$$\Sigma(k|k) = \Sigma(k|k-1) - H(k) S(k) H(k)^T$$

where $\Sigma(k|k) \triangleq \text{cov}\{x(k) | I(k)\}$

$H(k)$ is a "gain".

All the equations corresp. to the KALMAN FILTER.

Interpretation:

$$\hat{x}(k|k) = \hat{x}(k|k-1) + \underbrace{H(k) [y(k) - \hat{y}(k|k-1)]}_{\text{innovation}}$$

↑
estimation of $x(k)$
at inst. k
↑
Prediction
(w/o $y(k)$)
↑
actual $y(k)$
↑
predicted $y(k)$

Similar to Recursive ARX & Luenberger obs (but indices change)

When $y(k)$ and $\hat{y}(k|k-1)$ are very similar $\Rightarrow$ we can trust in the prediction.

Also, it is reasonable not to change the prediction when the output $y(k)$ is "NOT RELIABLE".

$$\underline{y}(k) = \underline{a}(k) \underline{x}(k) + \underline{\eta}(k)$$

In particular, if $\underline{\eta}(k)$ "dominates" $\Rightarrow \underline{y}(k)$ would be $\approx$ noise.

if $R(k) \rightarrow \infty \Rightarrow \underline{y}(k)$ would not be reliable.

$$\text{if } R(k) \rightarrow \infty \Rightarrow S(k) = \underline{a}(k) \Sigma(k|k-1) \underline{a}(k)^T + R(k) \rightarrow \infty$$

$$\Rightarrow S(k)^{-1} \rightarrow \underline{0} \Rightarrow H(k) = \Sigma(k|k-1) \underline{a}(k)^T S(k)^{-1} \rightarrow \underline{0}$$

Summing up.

$R(k) \rightarrow \infty \Rightarrow \hat{x}(k|k) = \hat{x}(k|k-1)$ ONLY PREDICTION!!!
(output is useless $\Rightarrow$ we do not use)

WALKING & R5

Think of you driving your car. Estimation of the position looking around. If you experience "dark" (e.g.: dark gallery)

$\Rightarrow$ No more information from "sensors" (eyes)

Mathematically this corresponds to $R(k) \rightarrow \infty \Rightarrow$ Go on ...

using prediction - covariance will enlarge. (due to noises) (1.35)

~~Put~~ Putting all together:

Before process starts (no measurements available)

$$\Rightarrow E\{\underline{x}(0) | \text{nothing}\}$$

↑
we can denote by $I(-1) = \text{a priori info.}$

$$\text{Note: } I(k) = \left\{ \begin{array}{c} \underline{y}(0) \dots \underline{y}(k) \\ \underline{u}(0) \dots \underline{u}(k) \end{array} \right\}$$

$$\Rightarrow I(0) = \left\{ \begin{array}{c} \underline{y}(0) \\ \underline{u}(0) \end{array} \right\}, I(-1) = \{\}$$

$$\left. \begin{aligned} \hat{\underline{x}}(0|-1) &= E\{\underline{x}(0) | I(-1)\} = \underline{\alpha} \\ \Sigma(0|-1) &= \text{cov}\{\underline{x}(0) | I(-1)\} = \Sigma \end{aligned} \right\} \text{a-priori information}$$

As the first measurement $\underline{y}(0)$ will be available

$$\hat{\underline{x}}(0|0) = E\{\underline{x}(0) | I(0)\} = \hat{\underline{x}}(0|-1) + H(0)[\underline{y}(0) - \hat{\underline{y}}(0|-1)]$$

$$\hat{\underline{y}}(0|-1) = G(0) \hat{\underline{x}}(0|-1) = G(0)\underline{\alpha}$$

at inst. 0
(k=0)

(I, 91)

$$\hat{y}(0|-1) = G(0) \underline{x}$$

- * $S(0) = G(0) \Sigma G(0)^T + R(0)$
- * $H(0) = \Sigma G(0)^T S(0)^{-1}$

$$\hat{x}(0|0) = \underline{x} + H(0) [y(0) - G(0) \underline{x}]$$

- * $\Sigma(0|0) = \Sigma - H(0) S(0) H(0)^T$

Next: assume we are at instant "k" (k > 0), when y(k) will be available)
 $\hat{x}(k-1|k-1)$, $\Sigma(k-1|k-1)$ already computed ($\Rightarrow$ AVAILABLE)

$$\hat{x}(k|k-1) = F(k-1) \hat{x}(k-1|k-1) + W(k-1) u(k-1)$$

- * $\Sigma(k|k-1) = F(k-1) \Sigma(k-1|k-1) F(k-1)^T + Q(k-1)$

$$\hat{y}(k|k-1) = G(k) \hat{x}(k|k-1)$$

- * $S(k) = G(k) \Sigma(k|k-1) G(k)^T + R(k)$

- * $H(k) = \Sigma(k|k-1) G(k)^T S(k)^{-1}$

$$\hat{x}(k|k) = \hat{x}(k|k-1) + H(k) [y(k) - \hat{y}(k|k-1)]$$

- * $\Sigma(k|k) = \Sigma(k|k-1) - H(k) S(k) H(k)^T$

~~KAL~~
KALMAN
FILTER

Remarks : •) different forms for $H(k)$, $\Sigma(k|k)$, $\neq S(k)$ (IIR) are available. Some make use of propagation of $S(k)^{-1}$ (w/o comput. of $S(k)$) (through Inversion Matrix Lemma)

-) If the structure of the system $F(k), W(k), G(k)$
 $Q(k), R(k) \quad k=0,1,\dots$

were available "a-priori" (in particular, ~~is~~ true in the time-invariant setting)

$\Rightarrow$ all the stored quantities can be computed A-Priori (all the matrices!) including the covariances.

- all the vectors $\wedge$ must be computed ON LINE
($k > 0$)

because they use measurements of outputs $\neq$ inputs (in particular, outputs are affected by disturbances)

I

- It can be shown that if the system is observable $\Rightarrow$ KF converges.

In particular $H(k)$, $\Sigma(k|k)$ converge.

In the time invariant setting one could use

$$H(\infty) = \lim_{k \rightarrow \infty} H(k) \text{ as a constant gain.}$$

$$\hat{x}(k|k) = \hat{x}(k|k-1) + H [y(k) - \hat{y}(k|k-1)]$$

~~one~~ Not necessary to ~~prop~~ propagate covariances

For estimations only.

Non linear case

$$\begin{cases} \underline{x}(k+1) = \underline{f}_k [\underline{x}(k), \underline{u}(k)] + \underline{\xi}(k) \\ \underline{y}(k) = \underline{g}_k [\underline{x}(k)] + \underline{\eta}(k) \end{cases}$$

$\underline{f}_k(\cdot)$, $\underline{g}_k(\cdot)$ be C^1 functions.

$$\Rightarrow \hat{\underline{x}}(k|k) = \hat{\underline{x}}(k|k-1) + H(k) [y(k) - \hat{y}(k|k-1)] \quad (I.99)$$

where:

For predictions
Using $\downarrow$
the nonlinear
functions.

$$\hat{\underline{x}}(k|k-1) = \underline{f}_{k-1} [\hat{\underline{x}}(k-1|k-1), \underline{u}(k-1)]$$

Attention to
indices

$$\hat{y}(k|k-1) = g_k [\hat{\underline{x}}(k|k-1)]$$

As for $H(k)$ we use the linearization of f_{k-1} and g_k around the estimates.

$$H(k) = \Sigma(k|k-1) G(k)^T S(k)^{-1}$$

$$\Sigma(k|k) = \Sigma(k|k-1) - H(k) S(k) H(k)^T$$

we need also

$$\Sigma(k|k-1) = F(k-1) \Sigma(k-1|k-1) F(k-1)^T + Q(k-1)$$

$$S(k) = G(k) \Sigma(k|k-1) G(k)^T + R(k)$$

with

$$F(k-1) \triangleq \frac{\partial \underline{f}_{k-1}(\underline{x}_{k-1}, \underline{u}_{k-1})}{\partial \underline{x}_{k-1}} \quad \left| \begin{array}{l} \underline{x}_{k-1} = \hat{\underline{x}}(k-1|k-1) \\ \underline{u}_{k-1} = \underline{u}(k-1) \end{array} \right.$$

$$G(k) \triangleq \left. \frac{\partial g_k(x_k)}{\partial x_k} \right|_{x_k = \hat{x}(k|k-1)} \quad \text{Note that } \hat{x}(k|k) \text{ is not yet available.} \quad (I_{200})$$

All this goes under the name of Extended Kalman Filter.

Remarks.

- Linear case we could compute matrices a priori in some case.

Here we need all the computations to be performed ON LINE

('cause of linearizations around estimates, that depends on on-line information).

- Linear KF is BLUE = Best Linear Unbiased Estimator
i.e. also if gaussianity does not hold true
 $\Rightarrow$ KF is OPTIMAL among linear Estimators

In the Gaussian case KF is OPTIMAL
linear &

(120)

- In the Non lin case EKF is "Reasonable".

Convergence of EKF can be dem. only in some particular situation (with assumptions).

- EKF can be use with state augmentation for estimation of parameters.

This gives also a quantification of quality of estim. (^{by} ~~through~~ covariances)

- Estimates and Cov. in EKF are only "Reasonable".

(Note cov. use linearization)

⇓
further approx.